

Physics Midterm · Practice Test 2 (Challenge)

Physics · increasing difficulty · key terms, careful calculations and diagram details

Name: _____

Date: _____

Period: _____

Time: 55 minutes. **Total:** 80 points. Part A: 18 multiple-choice questions, 2 points each (36). Part B: 5 free-response questions (44). The questions get harder as you go. This test is marked strictly: a missing unit, sign, direction, force label or key term loses the point.

Reference. $g = 9.81 \text{ m/s}^2$. Constant acceleration: $v = u + at$ · $s = ut + \frac{1}{2}at^2$ · $v^2 = u^2 + 2as$ · $s = \frac{1}{2}(u + v)t$. Components: $A_x = A \cos \theta$, $A_y = A \sin \theta$ (θ measured counter-clockwise from the +x axis). Newton's second law: $F_{\text{net}} = ma$. Weight: $W = mg$. Up and right are positive unless a question says otherwise. A calculator may be used (degree mode).

Part A · Multiple Choice (2 points each; circle one answer)

LEVEL 1 · WARM-UP

- Which quantity is NOT a vector?
A weight
C speed
B velocity
D acceleration
- A vector has components $A_x = 3.0 \text{ m}$ and $A_y = -4.0 \text{ m}$. Its magnitude is
A 5.0 m
C 7.0 m
B 1.0 m
D -5.0 m
- An object moves at constant velocity. Which statement must be true?
A No forces act on it.
B The forces on it are balanced.
C A net force acts in the direction of motion.
D It is on a frictionless surface.
- The newton, written in SI base units, is
A $\text{kg} \cdot \text{m/s}$
C $\text{kg}/(\text{m} \cdot \text{s}^2)$
B $\text{kg} \cdot \text{m}^2/\text{s}^2$
D $\text{kg} \cdot \text{m/s}^2$
- A car travels 150 km in 2.5 h. Its average speed is
A 60 km/h
C 16.7 km/h
B 375 km/h
D 152.5 km/h
- The acceleration of an object in free fall near the Earth's surface (no air resistance)
A depends on its mass
B increases as it falls
C is zero at the top of its path
D is 9.81 m/s^2 downward at every point

LEVEL 2 · READ CAREFULLY

7. A car traveling at 72 km/h brakes to rest in 4.0 s. What is its acceleration?
A -18 m/s^2 **B** -5.0 m/s^2
C $+5.0 \text{ m/s}^2$ **D** -0.20 m/s^2
8. A force of 10 N points at 120° from the +x axis. Its x-component is
A +5.0 N **B** +8.7 N
C -5.0 N **D** -8.7 N
9. A ball is thrown straight up at 15 m/s. Taking up as positive, what is its velocity 2.0 s later?
A -4.6 m/s **B** +4.6 m/s
C -34.6 m/s **D** +34.6 m/s
10. Stone 1 is thrown horizontally at 8.0 m/s from a bridge. Stone 2 is thrown horizontally at 16 m/s from the same height. Compared with stone 1, stone 2
A lands in half the time
B takes twice as long to land
C lands at the same time and the same distance out
D lands at the same time but twice as far out
11. Two forces act on a 2.0 kg object: 10 N east and 4.0 N west. Its acceleration is
A 7.0 m/s^2 east **B** 3.0 m/s^2 east
C 5.0 m/s^2 east **D** 3.0 m/s^2 west
12. A 60 kg student stands on a scale in an elevator that moves UP at a constant 2.0 m/s. The scale (normal force) reads
A 589 N **B** 709 N
C 469 N **D** 120 N

LEVEL 3 · CHALLENGE

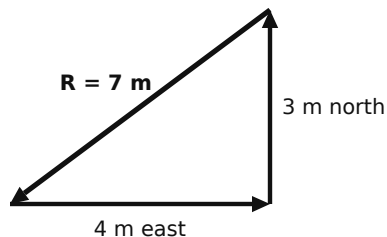
13. A ball is launched at 20 m/s at 30° above the horizontal. What is its speed at the highest point?
A 0 m/s **B** 10 m/s
C 20 m/s **D** 17.3 m/s
14. On level ground, two launch angles give the same range for the same launch speed. If one angle is 25° , the other is
A 50° **B** 75°
C 65° **D** 155°
15. A horse pulls a cart forward. The cart pulls back on the horse with an equal force. The cart can still accelerate because
A the horse's pull on the cart is slightly larger than the cart's pull on the horse
B the two forces act on different objects; only the forces on the cart decide the cart's acceleration
C Newton's third law does not apply to accelerating objects
D the horse pulls first, before the cart pulls back

- 16.** A ball has left the thrower's hand and is moving upward. Ignoring air resistance, the forces acting on the ball are
- A** gravity and the force of the throw
 - B** gravity only
 - C** gravity, the force of the throw, and inertia
 - D** no forces at the highest point
- 17.** Block A (4.0 kg) and block B (1.0 kg) are in contact on a frictionless floor. A 10 N horizontal force pushes on A, which pushes B. The force that A exerts on B is
- A** 10 N
 - B** 8.0 N
 - C** 5.0 N
 - D** 2.0 N
- 18.** A person walks 5.0 m at 0° , then 5.0 m at 90° , then 5.0 m at 180° (angles from the $+x$ axis). The resultant displacement is
- A** 5.0 m at 90°
 - B** 15 m at 90°
 - C** 7.1 m at 45°
 - D** 0 m

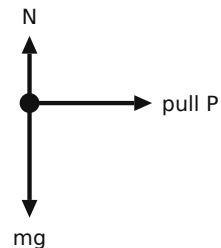
Part B · Free Response (show your work)

19. Find the mistakes (8 points)

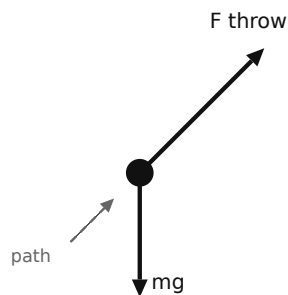
A student drew the four diagrams below. Each one contains at least one mistake. For each diagram, state every mistake and draw the correct diagram. (2 points each)



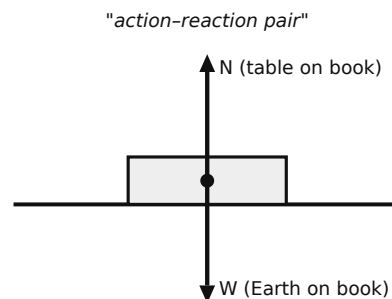
A. "4 m east then 3 m north"



B. crate pulled at constant velocity across a rough floor



C. ball moving up and to the right after leaving the hand (no air resistance)



D. book at rest on a table

A. Mistake(s) and correct diagram:

B. Mistake(s) and correct diagram:

C. Mistake(s) and correct diagram:

D. Mistake(s) and correct diagram:

20. Careful calculations (10 points)

(a) Find the x- and y-components of each vector. Angles are measured counter-clockwise from the +x axis. Give the correct sign and unit. (4 points)

Vector	x-component	y-component
25 m/s at 150°		
40 N at 250°		
12 m at 320°		
8.0 m/s^2 at 90°		

(b) Force A is 6.0 N at 0° . Force B is 8.0 N at 120° . Use components to find the magnitude and direction of the resultant $A + B$. (3 points)

Work:

(c) A stone is thrown straight up at 12 m/s from the edge of a ledge 20 m above the ground. Taking up as positive, find how long it takes to hit the ground. (3 points)

Work:

21. Projectile launched at an angle (10 points)

A soccer ball is kicked from level ground at 25 m/s at 40° above the horizontal. Ignore air resistance.

(a) Find the horizontal and vertical components of the launch velocity. (2 points)

Work:

(b) Find the time of flight. (2 points)

Work:

(c) Find the maximum height. (2 points)

Work:

(d) Find the range. (2 points)

Work:

(e) Find the magnitude and direction of the ball's velocity 1.0 s after it is kicked. (2 points)

Work:

22. Explain using the correct key terms (8 points)

(a) Passengers lurch forward when a bus brakes suddenly. Explain why. (2 points)

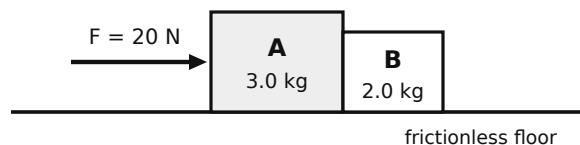
(b) A ball thrown straight up is momentarily at rest at its highest point. A student says its acceleration is zero there. Explain why the student is wrong. (2 points)

(c) In a collision between a truck and a small car, the car is damaged far more. A student says the truck must have hit the car with a bigger force. Explain what is really happening, using two of Newton's laws. (2 points)

(d) When you start to walk, which force pushes you forward? Name the third-law pair involved. (2 points)

23. Newton's second and third laws together (8 points)

Block A (3.0 kg) and block B (2.0 kg) are in contact on a frictionless floor. A horizontal force of 20 N pushes on block A, as shown.



(a) Draw a separate free-body diagram for each block. Label every force. (3 points)

Block A:

Block B:

(b) Find the acceleration of the blocks. (2 points)

Work:

(c) Find the size of the force that A exerts on B, and state the size and direction of the force that B exerts on A. (2 points)

Work:

(d) The 20 N force is now applied to B instead, pushing to the left. Is the contact force between the blocks the same as in (c)? Justify your answer with a calculation. (1 point)

Work: